Report A — Dataset Report

What our microwave data is, how it is stored as matrices, and the actual real / imaginary numbers (not graphs).

This report uses simple words, but it covers every dataset field the software uses. A dataset here is a collection of microwave measurements: frequencies plus complex S-parameters. Complex means each number has two parts: real and imaginary. Those two parts together describe both strength and delay of the wave. We print the numbers themselves.

Important: these files are research / educational. Output is not a medical diagnosis.

1. What a dataset is in this project

Think of a microwave scan as answering: "If I send a tiny radio wave into a phantom (a model of breast tissue), what comes back at each frequency?" The answer is not a picture yet. It is a table of complex numbers. The picture is made later by reconstruction. Every loader (MATLAB, Touchstone, UM-BMID) converts the file into one common object called **MicrowaveDataset**. Downstream modules never need to know the original file type.

- **file_path / file_name** — where the measurement lives on disk.
- **file_type** — MATLAB (legacy or v7.3), Touchstone (.s1p/.s2p/.s4p/.s8p), or MATLAB (UM-BMID).
- **frequencies** — a 1-D list of frequency points in hertz (Hz). Example: 1.0 GHz = 1,000,000,000 Hz.
- **s_parameters** — the complex matrix. Shape depends on file family (explained below).
- **n_ports** — how many physical ports / antennas. For BMID this is the antenna count.
- **available_variables** — names found in a .mat file, or S11,S21,… for Touchstone.
- **metadata** — extra facts: comments, antenna radius, wave speed, field of view, tumor labels.
- **raw** — optional extras such as the extracted S21 trace from a .s2p file.

The GUI "Dataset Information" panel shows a summary: file name, type, number of samples (how many complex numbers), number of frequencies, frequency range, number of ports, memory size, and variable names. Number of samples = every real+imag pair counted as one complex value.

2. Why numbers are complex (real and imaginary)

A microwave measurement is a sinusoid. You need amplitude (how strong) and phase (how delayed). Storing both as one complex number is the standard microwave way:

```
S = Re + j·Im
|S| = sqrt(Re² + Im²) (magnitude, linear)
angle(S) = atan2(Im, Re) (phase in radians)
|S|_dB = 20 log10(|S|)
```

The project does **not** throw away the imaginary part. Reconstruction needs phase, because delay-and-sum steering is a timing problem. If you only kept |S|, you could not form a focused image.

Touchstone files may write the same complex number in three formats. Our sample uses **RI** (real / imaginary). The other two are **MA** (magnitude / angle in degrees) and **DB** (dB / angle). The loader converts all of them to complex linear form before anything else runs.

```
DB → linear: |S| = 10^(dB/20), then S = |S| · (cos θ + j sin θ), θ in radians.
MA → complex: S = mag · (cos θ + j sin θ).
RI → complex: S = Re + j Im (already the form we want).
```

3. Two real data families we support

3.1 Touchstone .s2p (undergraduate S21 track)

A 2-port network analyzer file. At every frequency there is a full 2×2 scattering matrix. S11 is reflection at port 1, S21 is transmission from port 1 to port 2 (the usual imaging trace in the UG brief), S12 is the reverse transmission, S22 is reflection at port 2. In memory the array shape is **(n_freq, 2, 2)**.

3.2 MATLAB traces / UM-BMID cubes

University of Manitoba Breast Microwave Imaging Dataset (UM-BMID) stores many scans in one cube. Typical files: **fd_data_s21_adi.mat** (measurements) plus **md_list_s21_adi.mat** (labels). The cube is 3-D: **(n_scans, n_freq, n_antennas)** with n_freq = 1001 from 1 GHz to 8 GHz. Values are complex. One scan is sliced out as a 2-D matrix of shape **(1001, n_antennas)**. The suffix **_adi** means adipose-referenced / already cleaned, so Module 2 uses pass-through. **_emp** would be empty-chamber style calibration data. S11 vs S21 is in the file name.

The large UM-BMID cubes are gitignored (too big for git). If they are present under microwave/umbmid/simple-clean or project/datasets, the loader reads the real cube. This report therefore prints the actual numbers from the in-repo working files: sample_touchstone.s2p and sample_matlab.mat. The BMID layout is described with the exact axes and metadata fields the code expects.

4. Components of the Touchstone file (actual header)

File: **sample_touchstone.s2p**. Type: Touchstone (.s2p). Ports: 2. Frequencies: 201. Samples (complex entries): 804. Range: 1.0000–9.0000 GHz. S-array shape: (201, 2, 2) which means 201 frequencies × 2 × 2 complex matrix.

```
Option line parsed by the project:
frequency unit = GHZ
parameter type = S
data format = RI (RI = real, imaginary pairs)
reference impedance R = 50.0 ohm
start = 1.0000 GHz, stop = 9.0000 GHz, N = 201 samples
step Δf = 40.0000 MHz
```

Each data row in the file is one frequency followed by eight real numbers: Re(S11) Im(S11) Re(S21) Im(S21) Re(S12) Im(S12) Re(S22) Im(S22). That is the industry Touchstone order, not “row-major of the printed 2×2 from left to right”. The 2×2 matrix we reconstruct in memory is:

```
S(f) = [[ S11(f), S12(f) ],
[ S21(f), S22(f) ]]
```

Comments stored in metadata (physical hints the loader also parses): Synthetic 2-port Touchstone sample dataset Created with skrf 2.0.1 (http://scikit-rf.org). freq ReS11 ImS11 ReS21 ImS21 ReS12 ImS12 ReS22 ImS22

Parsed geometry defaults from those comments: antenna_radius_m = 0.08, wave_speed_m_per_s = 300000000.0, reconstruction spans = x (-0.05, 0.05), y (-0.05, 0.05).

5. Actual 2×2 matrices at selected frequencies

These are the true complex values from sample_touchstone.s2p. No plotting. Each entry is Re + j Im.

```
At 1.00 GHz, the full 2×2 S-matrix is
[  +0.192264 -0.174248j    +0.126712 -0.018170j  ]
[  +0.126712 -0.018170j    +0.241721 -0.127280j  ]
Order: row1 = S11, S12    row2 = S21, S22

At 2.00 GHz, the full 2×2 S-matrix is
[  -0.005221 -0.362314j    +0.093134 -0.164047j  ]
[  +0.093134 -0.164047j    +0.103205 -0.359478j  ]
Order: row1 = S11, S12    row2 = S21, S22

At 3.00 GHz, the full 2×2 S-matrix is
[  -0.356330 -0.324922j    -0.059404 -0.231544j  ]
[  -0.059404 -0.231544j    -0.261841 -0.442464j  ]
Order: row1 = S11, S12    row2 = S21, S22
```

```
At 4.00 GHz, the full 2×2 S-matrix is
[  -0.577092 +0.040319j    -0.246302 -0.150988j  ]
[  -0.246302 -0.150988j    -0.580795 -0.140807j  ]
Order: row1 = S11, S12    row2 = S21, S22

At 5.00 GHz, the full 2×2 S-matrix is
[  -0.395434 +0.442660j    -0.296220 +0.069993j  ]
[  -0.296220 +0.069993j    -0.533399 +0.347288j  ]
Order: row1 = S11, S12    row2 = S21, S22

At 6.00 GHz, the full 2×2 S-matrix is
[  +0.047698 +0.555927j    -0.144866 +0.248265j  ]
[  -0.144866 +0.248265j    -0.122466 +0.568730j  ]
Order: row1 = S11, S12    row2 = S21, S22

At 7.00 GHz, the full 2×2 S-matrix is
[  +0.375941 +0.295219j    +0.067114 +0.236057j  ]
[  +0.067114 +0.236057j    +0.274840 +0.414466j  ]
Order: row1 = S11, S12    row2 = S21, S22

At 8.00 GHz, the full 2×2 S-matrix is
[  +0.358609 -0.048829j    +0.171645 +0.087291j  ]
[  +0.171645 +0.087291j    +0.388770 +0.070873j  ]
Order: row1 = S11, S12    row2 = S21, S22

At 9.00 GHz, the full 2×2 S-matrix is
[  +0.166382 -0.197305j    +0.105671 -0.036646j  ]
[  +0.105671 -0.036646j    +0.203992 -0.149112j  ]
Order: row1 = S11, S12    row2 = S21, S22
```

6. Full numeric table — first 20 frequency rows

Every number below is copied from the file. Columns are real and imaginary parts. S12 equals S21 in this synthetic reciprocal sample (you can check the digits).

f GHz	Re S11	Im S11	Re S21	Im S21	Re S12	Im S12	Re S22	Im S22
1.00	0.192264	-0.174248	0.126712	-0.018170	0.126712	-0.018170	0.241721	-0.127280
1.04	0.143513	-0.183659	0.118527	-0.026081	0.118527	-0.026081	0.231808	-0.126199
1.08	0.170106	-0.197048	0.127938	-0.037581	0.127938	-0.037581	0.219484	-0.151563
1.12	0.156737	-0.219077	0.127871	-0.038371	0.127871	-0.038371	0.240202	-0.141636
1.16	0.154007	-0.204091	0.126494	-0.042905	0.126494	-0.042905	0.238996	-0.160248
1.20	0.152120	-0.237926	0.133137	-0.052777	0.133137	-0.052777	0.232940	-0.175435
1.24	0.129440	-0.222473	0.122814	-0.053450	0.122814	-0.053450	0.229050	-0.196594
1.28	0.142301	-0.237730	0.131469	-0.055779	0.131469	-0.055779	0.230348	-0.191000
1.32	0.130564	-0.234839	0.122286	-0.059338	0.122286	-0.059338	0.221446	-0.196757
1.36	0.166653	-0.255219	0.123087	-0.060895	0.123087	-0.060895	0.220570	-0.231588
1.40	0.129500	-0.271108	0.121425	-0.072272	0.121425	-0.072272	0.201406	-0.212289
1.44	0.117144	-0.267512	0.128561	-0.079981	0.128561	-0.079981	0.220159	-0.218011
1.48	0.110898	-0.284774	0.121042	-0.080962	0.121042	-0.080962	0.203001	-0.241369
1.52	0.099682	-0.288907	0.125207	-0.091475	0.125207	-0.091475	0.216020	-0.250067
1.56	0.088076	-0.294435	0.118260	-0.099974	0.118260	-0.099974	0.170003	-0.241028
1.60	0.086600	-0.303765	0.128061	-0.093417	0.128061	-0.093417	0.191683	-0.262955
1.64	0.086828	-0.311668	0.130481	-0.106482	0.130481	-0.106482	0.188201	-0.294272
1.68	0.070748	-0.325239	0.120782	-0.109424	0.120782	-0.109424	0.168268	-0.295372
1.72	0.073464	-0.325698	0.125998	-0.112238	0.125998	-0.112238	0.169085	-0.306557
1.76	0.052276	-0.351822	0.118490	-0.126476	0.118490	-0.126476	0.164460	-0.316131

7. Full numeric table — last 10 frequency rows

f GHz	Re S11	Im S11	Re S21	Im S21	Re S12	Im S12	Re S22	Im S22
8.64	0.242881	-0.167410	0.145678	-0.003459	0.145678	-0.003459	0.293360	-0.097931
8.68	0.216793	-0.167978	0.145569	-0.003843	0.145569	-0.003843	0.285978	-0.123337
8.72	0.204410	-0.182367	0.143038	-0.009933	0.143038	-0.009933	0.259911	-0.141745
8.76	0.197687	-0.185172	0.137222	-0.007925	0.137222	-0.007925	0.264679	-0.126286
8.80	0.183668	-0.194121	0.118594	-0.014425	0.118594	-0.014425	0.253627	-0.140934
8.84	0.187832	-0.174028	0.124738	-0.025553	0.124738	-0.025553	0.248257	-0.127874
8.88	0.187516	-0.188638	0.130905	-0.020002	0.130905	-0.020002	0.249111	-0.128666
8.92	0.169731	-0.197354	0.119355	-0.025834	0.119355	-0.025834	0.225609	-0.138239
8.96	0.165279	-0.207595	0.115460	-0.037321	0.115460	-0.037321	0.212693	-0.149227
9.00	0.166382	-0.197305	0.105671	-0.036646	0.105671	-0.036646	0.203992	-0.149112

8. Magnitude and phase of S21 (still numbers, not graphs)

Module 2's UG track extracts only S21. Here are |S21| and phase at the same selected frequencies. Phase is unwrapped later in preprocessing; here you see the raw wrapped angle.

f GHz	Re S21	Im S21	|S21| linear	|S21| dB	phase deg (wrapped)
1.00	0.126712	-0.018170	0.128008	-17.855	-8.160
2.00	0.093134	-0.164047	0.188640	-14.487	-60.415
3.00	-0.059404	-0.231544	0.239043	-12.430	-104.389
4.00	-0.246302	-0.150988	0.288898	-10.785	-148.491
5.00	-0.296220	0.069993	0.304376	-10.332	166.706
6.00	-0.144866	0.248265	0.287440	-10.829	120.264
7.00	0.067114	0.236057	0.245412	-12.202	74.129
8.00	0.171645	0.087291	0.192566	-14.308	26.956
9.00	0.105671	-0.036646	0.111845	-19.028	-19.126

9. MATLAB 8-antenna matrix (sample_matlab.mat)

This file is a synthetic UWB array: frequencies 1.0–9.0 GHz, 201 points, S-parameter shape (201, 8) meaning (n_freq, n_traces). Each column is one antenna trace. All values are complex. n_ports reported by the loader for trace-organized MATLAB is 1 (for this style the second axis is traces; reconstruction treats columns as antennas).

Variables in the file: frequency, s_parameters, notes. Frequency variable / S variable from metadata: frequency, s_parameters. Frequencies were stored in GHz inside the .mat file; the loader multiplies by 1e9 because max(frequency) < 1000, which is the project's GHz→Hz heuristic.

At 1.00 GHz, the 1×8 complex row (one number per antenna) is:

```
ant 0: +0.175311 -0.156748j
ant 1: +0.219372 -0.118575j
ant 2: +0.262676 -0.062612j
ant 3: +0.251081 +0.024529j
ant 4: +0.229812 +0.106908j
ant 5: +0.166789 +0.144615j
ant 6: +0.129884 +0.205472j
ant 7: +0.050023 +0.246567j
```

At 5.00 GHz (middle of the sweep), the 1×8 complex row is:

```
ant 0: -0.398775 +0.452591j
ant 1: -0.540756 +0.309174j
ant 2: -0.598004 +0.151916j
ant 3: -0.603866 -0.023354j
ant 4: -0.551322 -0.200158j
ant 5: -0.451970 -0.345654j
ant 6: -0.319303 -0.460772j
```

```
ant 7: -0.188375 -0.554562j
```

At 9.00 GHz, the 1×8 complex row is:

```
ant 0: +0.168345 -0.201290j
ant 1: +0.227286 -0.165493j
ant 2: +0.220357 -0.075586j
ant 3: +0.249428 -0.004083j
ant 4: +0.231237 +0.087253j
ant 5: +0.171112 +0.120543j
ant 6: +0.142307 +0.175930j
ant 7: +0.082266 +0.223356j
```

9.1 Real-part matrix and imaginary-part matrix

The same data can be shown as two real matrices of shape (n_freq × 8). First 8 frequency rows of the REAL part:

f GHz	A0 Re	A1 Re	A2 Re	A3 Re	A4 Re	A5 Re	A6 Re	A7 Re
1.000	0.17531	0.21937	0.26268	0.25108	0.22981	0.16679	0.12988	0.05002
1.040	0.17729	0.24390	0.26000	0.25481	0.23902	0.18906	0.12425	0.07376
1.080	0.16923	0.20758	0.25997	0.26754	0.23128	0.21695	0.12817	0.07940
1.120	0.14938	0.22838	0.25056	0.26331	0.25149	0.21749	0.15769	0.08717
1.160	0.16759	0.23100	0.26824	0.26685	0.25690	0.22800	0.15612	0.10884
1.200	0.15874	0.22380	0.28160	0.27341	0.24360	0.21790	0.18171	0.09541
1.240	0.14427	0.22990	0.25267	0.28806	0.24564	0.23828	0.18267	0.10840
1.280	0.15043	0.20571	0.26581	0.28636	0.27861	0.26926	0.18053	0.12903

First 8 frequency rows of the IMAGINARY part:

f GHz	A0 Im	A1 Im	A2 Im	A3 Im	A4 Im	A5 Im	A6 Im	A7 Im
1.000	-0.15675	-0.11857	-0.06261	0.02453	0.10691	0.14462	0.20547	0.24657
1.040	-0.19626	-0.12875	-0.04350	0.04952	0.08463	0.15888	0.17967	0.22725
1.080	-0.19057	-0.12281	-0.08234	0.00189	0.09412	0.12815	0.19362	0.23195
1.120	-0.20973	-0.15132	-0.08256	-0.01781	0.09888	0.12379	0.19478	0.23012
1.160	-0.21192	-0.16397	-0.08253	0.00531	0.08454	0.13360	0.20052	0.22013
1.200	-0.23302	-0.18489	-0.10847	-0.02509	0.04656	0.14988	0.19461	0.22502
1.240	-0.21033	-0.18585	-0.10265	-0.02845	0.06494	0.14452	0.19042	0.23983
1.280	-0.24728	-0.16896	-0.13457	-0.03586	0.05420	0.11686	0.19096	0.22941

10. UM-BMID cube — how it is represented

When a real UM-BMID file is present, the loader looks for a 3-D numeric array whose length-1001 axis is frequency. It reorders axes so the working cube is always:

```
cube[scan_index, frequency_index, antenna_index] ∈ ■
scan slice S = cube[k, :, :] shape (1001, n_antennas)
frequencies_hz = linspace(1e9, 8e9, 1001)
```

Example of the idea (tiny toy numbers, only to show layout — not a real UM-BMID scan). Suppose 2 frequencies and 3 antennas for scan 0:

```
S = [[ 0.12-0.04j , 0.08+0.01j , 0.07-0.02j ],
[ 0.10-0.06j , 0.09+0.00j , 0.06-0.03j ]]
Row = frequency, column = antenna. Each cell is Re + j Im.
```

Companion metadata md_list_*.mat is a list of MATLAB structs, one per scan, with fields the code actually reads:

- **id** — scan identifier.
- **phant_id** — phantom name.
- **tum_diam** (cm) or **tum_rad** (radius; diameter = 2×radius if diameter missing).

- **tum_x, tum_y** (cm) — labeled tumor coordinates. A scan is marked tumor if diameter > 0 or a non-zero (x,y) is present.
- **tum_shape** — shape string when provided.
- **birads** — BIRADS bucket used only as a research label, not a diagnosis.
- **ant_rad** (cm) — antenna array radius; default 18 cm if missing → 0.18 m in reconstruction.

Those labels are copied into MicrowaveDataset.metadata as SI units: tumor_diameter_m, tumor_x_m, tumor_y_m, antenna_radius_m, dataset_family='UM-BMID', bmid_scan_index, bmid_has_tumor, bmid_sparam, bmid_calibration, reconstruction spans ±6 cm. Taxonomy then buckets size (small ≤ 2 cm, medium ≤ 4 cm, else large), severity from size/BIRADS, and quadrant from (x,y) with a 1 cm central disk.

11. Validation rules applied at load time

- Supported extensions: .mat, .s1p, .s2p, .s4p, .s8p.
- Frequencies must be finite and S-parameters finite (no NaN/Inf).
- Touchstone option line must declare S parameters and R (reference impedance).
- UM-BMID multi-scan cubes refuse to load until a scan index is chosen.
- Metadata-only files (md_list_*, metadata_*) are not measurements; automation pairs them to fd_data_*.
- corrupted.mat in datasets/ is an intentional broken file used to test error handling.

12. What is NOT in the dataset

The raw file is not an image. It has no pixels of a tumor. It has no diagnosis. Tumor x/y in BMID metadata is a laboratory label used later to score localization error. Touchstone sample_touchstone.s2p is synthetic (built by datasets/generate_sample_data.py) so that students can run the full app without a VNA. Real lab .s2p files use the same matrix layout.

Tumour classification status

Official first-scan score: **71/100**. This is the nested test on UM-BMID (signal plus image, extra training scans from phantoms that are not in the test fold, cut-off chosen on training phantoms only). It has not been replaced.

Official follow-up score: **97/100**. The scan is compared with earlier healthy scans of the same breast from the same sitting. Tumour and healthy cases are given the same number of those scans, at most 3, so the count cannot give the answer away.

A later side check scored about **69/100**. That check used only the signal features and only the fair training scans. Adding scattering-spectrum and frequency-slope formulas did not raise it. It is recorded as a side check, not as a new official score.

A convolutional network on the breast-window sinogram, and a PCA-then-logistic classifier on that same sinogram, were then scored on the fair phantom-held-out test. They scored **50/100** and **45/100**. The network could fit the breasts it trained on and did not carry that fit to a new phantom. Neither replaced the official first-scan model.

Two write-ups live in results/reports/classification/: **how_classification_works.pdf** (process, cut-offs and formulas, in plain language) and **guide_progress.pdf** (numbers and progress for a project guide). Machine-readable scores stay in results/classification/final_summary.json.

End of Report A. Report C explains how these matrices become an image. Report B explains how many files can be processed without clicking the GUI.